# Kanban and flow: WIP, lead time, cycle time

*your board becomes a measuring instrument*

DIG 123 · Agile Methodologies for Innovation · Class 6 · Monday, October 26, 2026 · Sprint 2 starts today




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From the Industry to the Classroom — Building the IT's Future, Today and Together !

# Where we are in the course

- Sprint 1: October 12 → 25 — Review and Retrospective done, real velocity counted (Class 5)
- Sprint 2: October 26 → November 8 — starts today, with a board that has limits
- Today's studio: CQ2 (Checkpoint Quiz 2) in the first 15 minutes — Classes 3 to 5, one velocity item, one burndown item
- No new lab today · Lab 04 'Flow & Digital Business Brief' opens November 2, feedback November 8
- Mid-Term, November 16: Part C uses today's table (lead / cycle time, throughput, Little's Law, a WIP limit to recommend); Part D asks Scrum or Kanban for a scenario
- All labs: final deadline Sunday, December 20, 23:59

# By the end of this hour you can…

- Explain what a kanban is (a card that orders replenishment) and how the idea moved to knowledge work in 2004–2010
- Name the parts of a Kanban board and of a Definition of Workflow; state the pull rule
- Define WIP (Work In Progress) and show with arithmetic why starting everything makes everything late
- Write a WIP limit as a policy on a column and say what the team does when the column is full
- Compute lead time, cycle time, work item age and throughput from the dates on the cards
- Apply Little's Law — average cycle time = average WIP ÷ throughput — and name its conditions
- Read a cumulative flow diagram; write a provisional Service Level Expectation
- Compare Scrum, Kanban and Scrumban and choose one for a kind of work, in five sentences with reasons

# The hour, minute by minute

| Time | Block | What happens |
|---|---|---|
| **0:00–0:02** | Check-in | Sprint 1 closed? Velocity counted? Three hands. |
| **0:02–0:12** | Lesson 1 | The board and the pull — Toyota, the Kanban Method, the Kanban Guide, the Definition of Workflow |
| **0:12–0:17** | Activity 1 | A wall board: Team Menu's cards on October 15 |
| **0:17–0:27** | Lesson 2 | WIP and WIP limits — the arithmetic of starting everything |
| **0:27–0:32** | Activity 2 | The multitasking demo (three names, two rounds) |
| **0:32–0:42** | Lesson 3 | Lead time, cycle time, throughput, Little's Law |
| **0:42–0:47** | Activity 3 | Fun-quiz ×3, computational |
| **0:47–0:57** | Lesson 4 | The cumulative flow diagram; Scrum vs. Kanban vs. Scrumban |
| **0:57–1:00** | Activity 4 | Flash-cards ×6 + 60-second zen break |

# Before we start: licence, sources, fictional data

- This lecture is educational material of Cyber.SoHo, licensed to you personally for your study (Legal Notice §4.2) — no copying, posting or feeding into machine-learning systems (§4.3)
- Definitions come from free public sources: the Kanban Guide 2025.5 (Coleman & Vacanti, CC BY-SA 4.0), Kanban University's guide to the Kanban Method, the Scrum Guide 2020 (CC BY-SA 4.0)
- Team Menu — Amira, Noah, Sofia, Karim — and its data are fictional (§20); your own boards must hold no personal data about outsiders
- Every external link and video was checked on October 23–24, 2026; open links at your own discretion and check them with VirusTotal (§14, Annex C)
- No recording of the session (§17) · academic integrity (§18)
- Full text: cyber-soho.github.io/prevmon/00.01. cyber.soho - legal.html

# 1

# The board and the pull

0:02 – 0:12 · then Activity 1 (0:12 – 0:17)

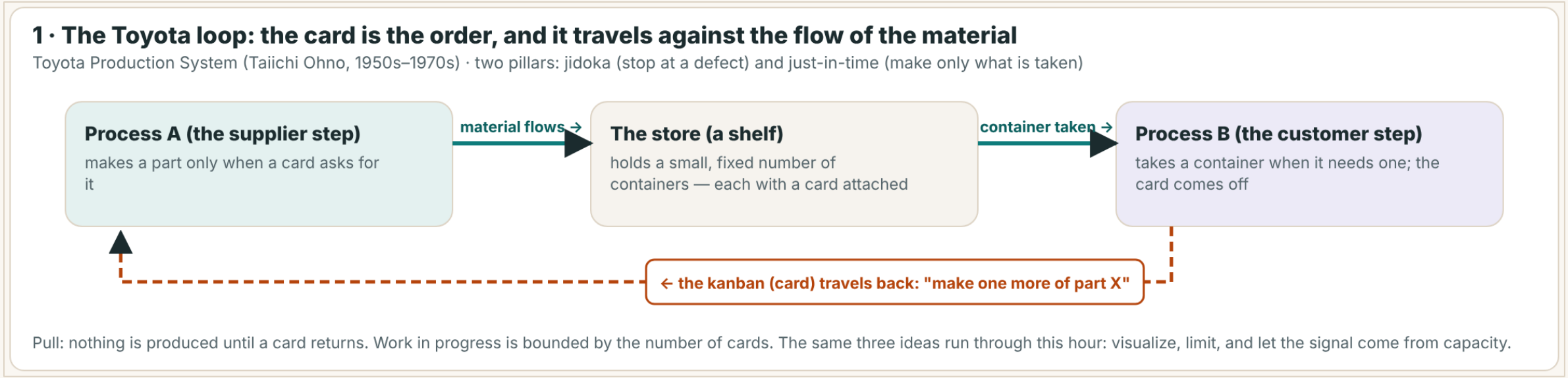
**BY THE END OF THIS LESSON YOU CAN**

Explain what a kanban is in the Toyota Production System — a card that orders replenishment when something is used up — and what changed when the idea moved to knowledge work in 2004–2010.

Name the parts of a Kanban board and of its Definition of Workflow: work items, states from started to finished, explicit policies, WIP control, the commitment point and the delivery point.

State the pull rule — a card enters a state only when there is a clear signal of capacity — and tell a pull from a push on your own board.

# A card on a factory shelf (1/2) · the Toyota loop




- A kanban ( 看板 , 'signboard' or 'card') travels back from the step that consumed a part to the step that makes it
- The card is the order: nothing is produced until a card returns
- Work in progress is bounded by the number of cards in circulation

# A card on a factory shelf (2/2) · what the card changed

- Toyota Production System (Taiichi Ohno, 1950s–1970s): two pillars — jidoka (stop at a defect) and just-in-time (make only what is taken)
- Push: a plan decides when to start; piles form between steps
- Pull: the next step's consumption decides; the card is the signal of capacity
- Three ideas that survive the move to knowledge work: visualize the work · limit what is open · let the signal come from capacity
- Sources: Toyota's own history pages; Ohno, 'Toyota Production System' (1978 / English 1988)

# From the factory to knowledge work · the timeline




- 2004 — David J. Anderson's first pull system for a Microsoft maintenance team (XIT)
- 2007 — Corbis: the practices named 'the Kanban Method' · 2010 — the book 'Kanban: Successful Evolutionary Change'
- 2008–09 — Corey Ladas: Scrumban · 2018 / 2021 — Kanban Guide for Scrum Teams (Scrum.org)
- 2020 → 2025.5 — the Kanban Guide (Coleman & Vacanti): three practices, four measures, a Definition of Workflow

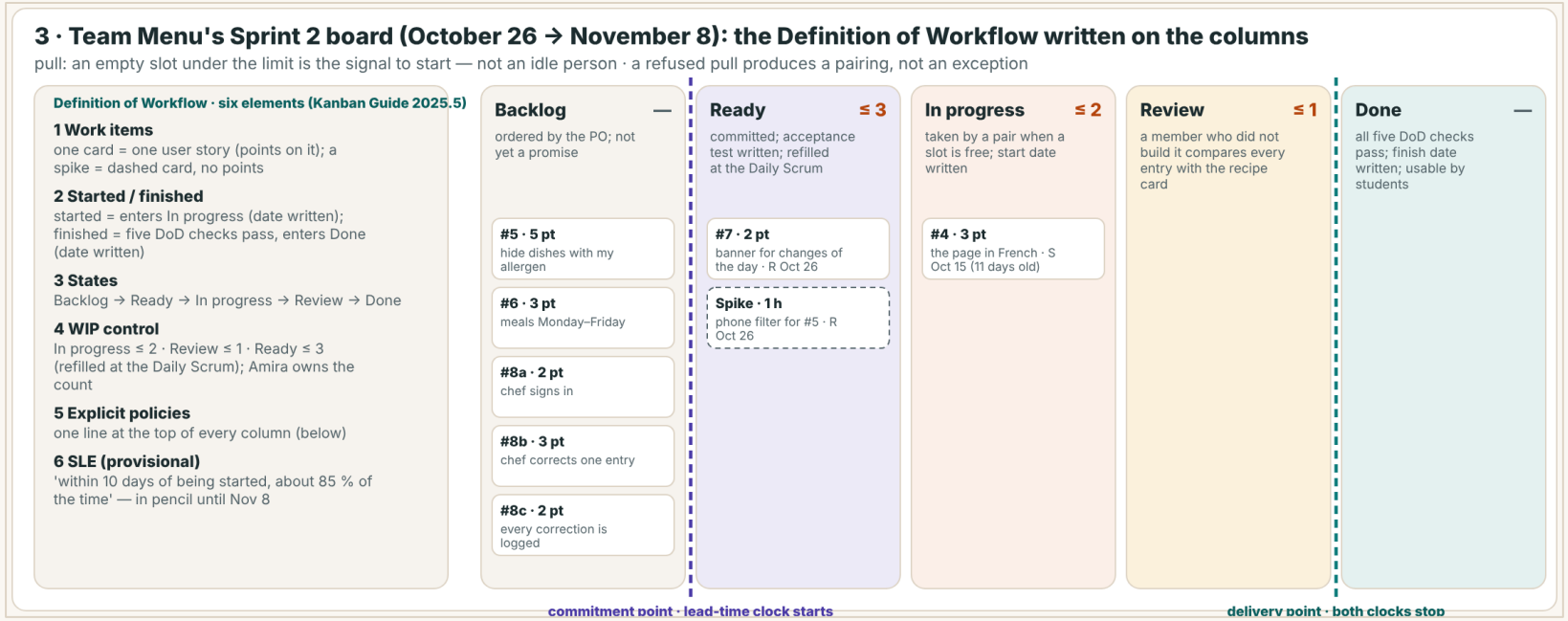
# Two free guides, one vocabulary

| | Kanban Method (Kanban University) | Kanban Guide 2025.5 (kanbanguides.org) |
|---|---|---|
| **Six general practices** | visualize · limit WIP · manage flow · make policies explicit · implement feedback loops · improve collaboratively, evolve experimentally | — |
| **Three practices** | — | defining and visualizing a workflow · actively managing items in a workflow · improving a workflow |
| **Measures** | lead time, throughput, WIP, flow efficiency, … | WIP · throughput · work item age · cycle time (the four of this course) |
| **Written definition of how work flows** | policies, classes of service, commitment | the Definition of Workflow (DoW): six required elements |
| **Promise to the customer** | service level agreement / expectation | the Service Level Expectation (SLE): a period and a probability |

# The Definition of Workflow · six required elements (Kanban Guide 2025.5)

- 1 A definition of the individual units of value moving through the workflow — the work items
- 2 A definition of when work items are started and finished
- 3 One or more defined states the items flow through, from started to finished
- 4 A definition of how WIP will be controlled, from started to finished
- 5 Explicit policies about how items can flow through each state
- 6 A Service Level Expectation — a forecast of how long an item should take, with a probability
- Written on the board, visible to everyone; changed by decision, not by habit

# Team Menu's Sprint 2 board · limits, policies, two points




- Backlog → Ready (≤ 3) → In progress (≤ 2) → Review (≤ 1) → Done
- Commitment point before Ready: the lead-time clock starts · Delivery point before Done: both clocks stop
- A policy at the top of every column; Amira owns the count

# The pull rule

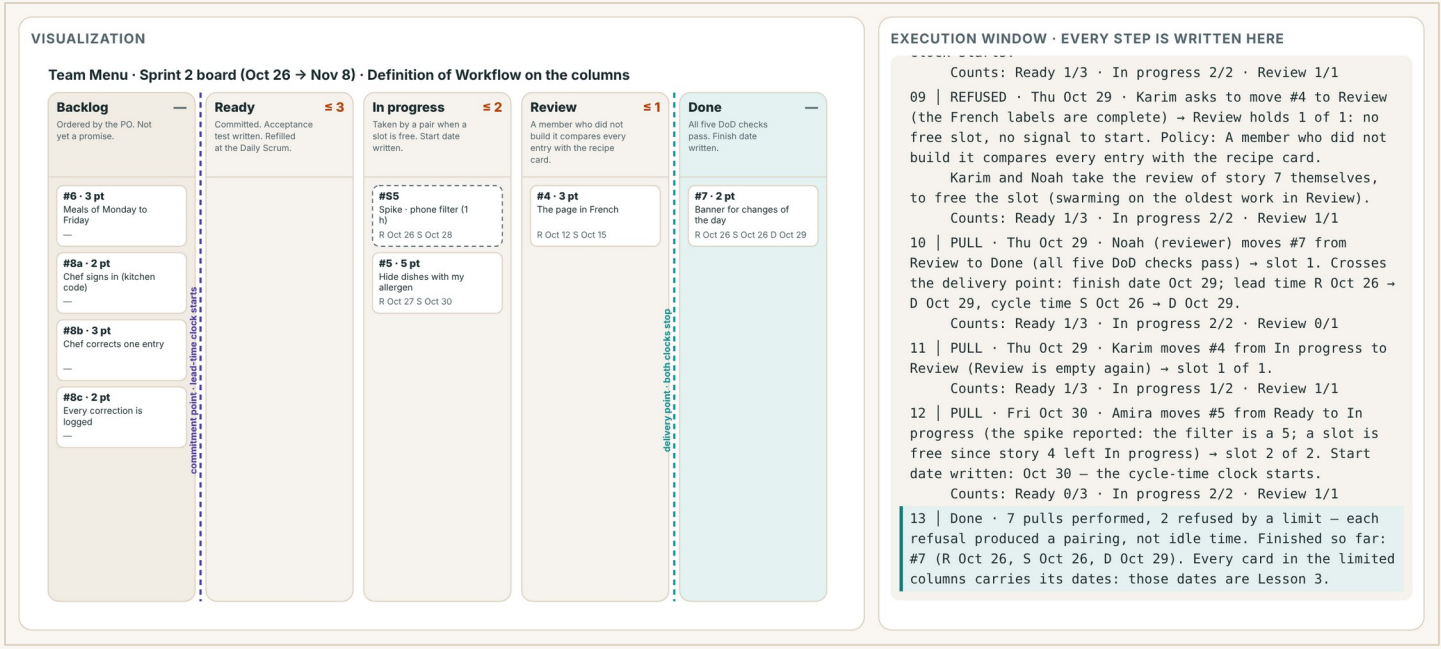
- A card enters a state only when there is a clear signal of capacity — an empty slot under the limit
- Not: an idle person · not: a manager's assignment · not: the date on a plan
- A refused pull produces a conversation, not an exception: pair on the oldest card, take a review, remove a blocker
- Push shows itself as piles between columns; pull shows itself as empty slots and short queues
- On your board: the question 'may I start?' is answered by the column, not by the person

# Activity 1 · a wall board (5 minutes)

ACTIVITY · EVERYONE TAKES PART

- Four columns on the wall: Ready · In progress · Review · Done
- Team Menu's Sprint 1 cards as they were on Thursday, October 15 (day 4): four stories, fourteen work items
- 2 min — place every card where it was that day; write the date it entered the column in its corner
- 2 min — count the cards per column; name the fullest column and the one empty the longest
- 1 min — one sentence: if nothing changes, where does the work pile up, and who waits?
- Keep a photo: in the studio this becomes your Sprint 2 board, with limits

# Example 1 · the board as a program: pulls accepted and refused




- Team Menu's five columns with limits and policies; eight cards of October 26 with their dates
- Nine pull requests of the first week checked against the limits: 7 performed, 2 refused — each refusal becomes a pairing
- Every accepted move writes a date on the card: those dates are Lesson 3's data

2

# WIP and WIP limits

0:17 – 0:27 · then Activity 2 (0:27 – 0:32)

**BY THE END OF THIS LESSON YOU CAN**

Define WIP with the Kanban Guide's words — the number of work items started but not finished — and count it on a board.

Show with arithmetic, before any switching cost, why starting everything at once makes every item late and delays the first feedback.

Write a WIP limit as an explicit policy on a column, say what the team does when the column is full, and explain why the limit is a signal to talk and not a quota.

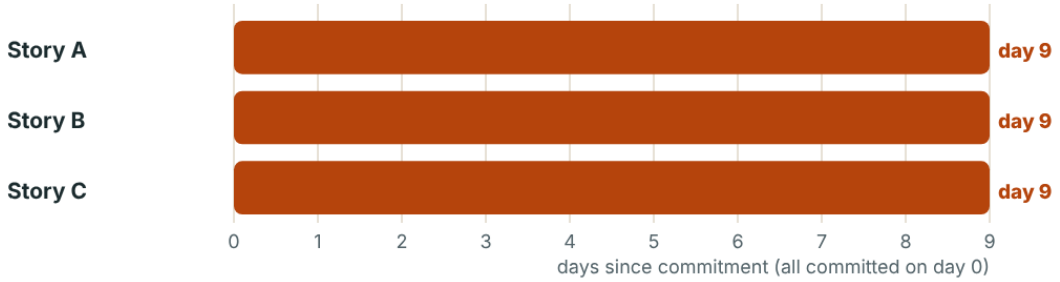
# What WIP is, and what it costs

- WIP (Work In Progress): the number of work items started but not finished (Kanban Guide 2025.5)
- Count it on the board: every card between the commitment point and the delivery point that has been started
- Team Menu on October 15: four stories started, none finished — WIP 4, the whole Sprint content open at once
- Open work costs: waiting customers, context switching, half-finished pieces that age, feedback that arrives late
- The question of this lesson is not 'how fast do we work?' but 'how much do we keep open?'

# Why starting everything makes everything late (1/2) · three equal stories

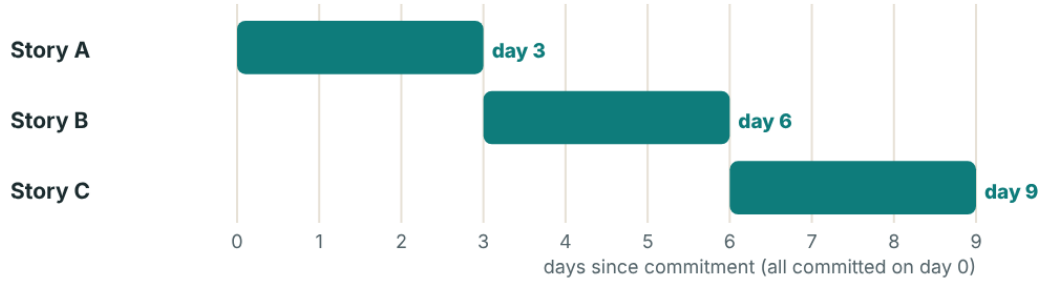
## 1 · Three equal stories of 3 days each — two policies, nine days of work either way

### A · start all three at once (equal split, 1/3 of the day each)




**average lead time 9.0 d · first delivery day 9**

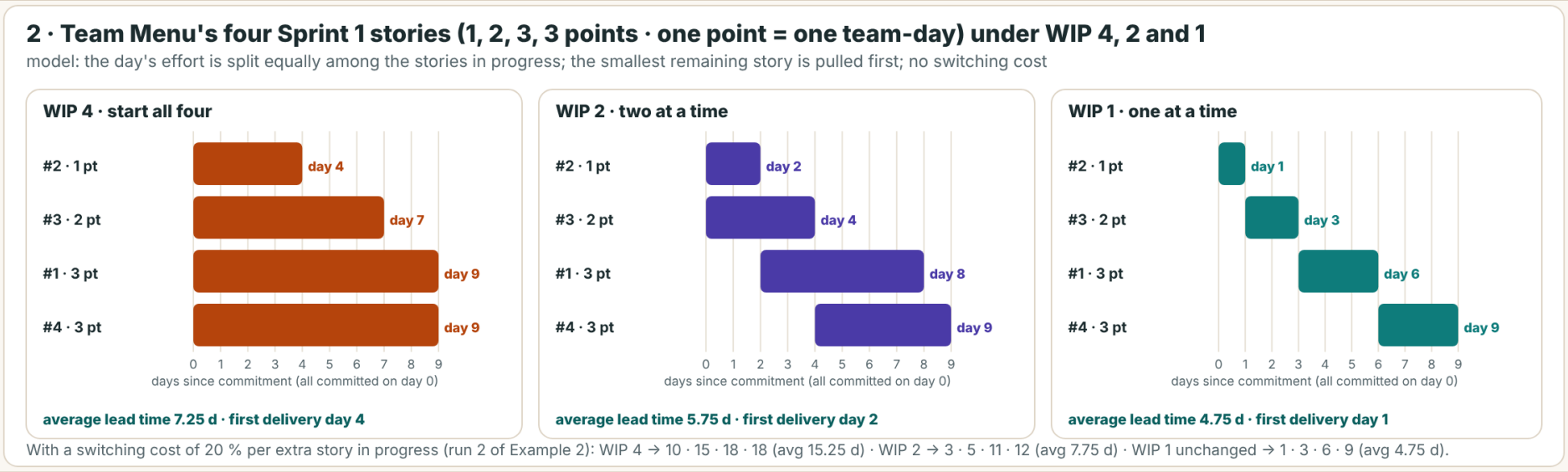
### B · one at a time (limit 1)




**average lead time 6.0 d · first delivery day 3 · a third less waiting**

- Three stories of three days each; the team has one unit of effort per day
- Start all three at once (equal split): every story is finished on day 9 — lead times 9, 9, 9; average 9; first delivery day 9
- One at a time: days 3, 6, 9 — average 6; first delivery day 3 — a third less waiting, with no switching cost counted yet

# Why starting everything makes everything late (2/2) · Team Menu's four stories




- Model: one point = one team-day; equal split; smallest remaining story first
- WIP 4 → finish days 4 · 7 · 9 · 9 — average lead time 7.25 d, first finished story on day 4
- WIP 2 → 2 · 4 · 8 · 9 — average 5.75 d, first on day 2 · WIP 1 → 1 · 3 · 6 · 9 — average 4.75 d, first on day 1
- The last story finishes on day 9 under every policy: a limit does not make the team faster, it stops the small stories waiting behind the large ones

# The switching cost · an assumption, stated

- Every switch between unfinished tasks costs time to re-enter the context
- American Psychological Association, summary of the task-switching experiments (Rubinstein, Meyer & Evans, 2001): up to 40 % of productive time lost for complex tasks
- Run 2 of Example 2: 20 % per extra story in progress — WIP 4 → 10 · 15 · 18 · 18 (average 15.25 d); WIP 2 → 3 · 5 · 11 · 12 (7.75 d); WIP 1 unchanged (4.75 d)
- The direction of the result does not depend on the size of the cost; only the distance between the policies does
- Write the assumption before the number — Lab 04 asks for exactly that

# The WIP limit · a policy written on a column

### 3 · The limit is a written policy on the column — and what to do when the column is full

**In progress ≤ 2**

Policy written at the top of the column: a card enters only when a slot under the limit is free. The count is owned by Amira and checked at every Daily Scrum. Review ≤ 1 · Ready ≤ 3.

**Team Menu, Sprint 1 Retrospective → in force from October 26**

**When the column is full: Pair**

join the oldest card in the column (swarm) — two people, one card, finished sooner

**When the column is full: Review**

take a card in Review: a finished review frees a slot downstream

**When the column is full: Unblock**

find what the oldest card is waiting for — a decision, data, a person — and remove it

Why a full column must not mean 'start something else': every switch between unfinished tasks costs time to re-enter the context. The American Psychological Association's summary of the switching-cost experiments (Rubinstein, Meyer & Evans, 2001) puts the loss at up to 40 % of productive time for complex tasks. The slogan is Arne Roock's (2012): stop starting, start finishing.

- 'In progress ≤ 2' at the top of the column: a card enters only when a slot under the limit is free
- When the column is full: pair on the oldest card · take a review · remove a blocker — not 'start something else'
- The limit is a signal to talk, not a quota; Arne Roock (2012): stop starting, start finishing

# Choosing a first limit — and Team Menu's limit

- Start from what the board shows: count the cards in each column over the last Sprint, set the limit a little below the usual count, adjust at the Retrospective
- A common first rule: about one or two items per pair of people in the column (Atlassian's guidance); never a limit so high that it is never reached
- Team Menu (Sprint 1 Retrospective): In progress ≤ 2 · Review ≤ 1 · Ready ≤ 3, refilled at the Daily Scrum; Amira owns the count
- In force from October 26; reviewed at the Retrospective of November 8
- The justification is the table of slide 19 — the same justification Part C of the Mid-Term asks for

# Activity 2 · the multitasking demo (5 minutes)

ACTIVITY · EVERYONE TAKES PART

- Pairs; one writes, one times. Three names of six letters: KANBAN · TOYOTA · REVIEW
- Round 1, interleaved: first letter of name 1, first of name 2, first of name 3, second of name 1 … Time it; count mistakes
- Round 2, one at a time: name 1 complete, then name 2, then name 3. Time it; count mistakes
- Compute: time of round 1 ÷ time of round 2 — and the moment the first name was finished in each round
- Eighteen letters in both rounds: the work was identical

# Example 2 · the WIP policy simulator




- Six simulations: WIP 4, 2 and 1, without and with a switching cost; one step = one simulated day
- Four progress bars per panel, every finish day on a timeline, average lead time and first finish updated live
- Change the four stories in the copied code and read your own three averages

3

# Lead time, cycle time, throughput, Little's Law

0:32 – 0:42 · then Activity 3 (0:42 – 0:47)

**BY THE END OF THIS LESSON YOU CAN**

Compute lead time, cycle time and work item age from the three dates on a card, in elapsed calendar days, and always answer the question 'from when?'.

Compute throughput as a count of finished items per unit of time, and tell it apart from velocity.

Apply Little's Law — average cycle time = average WIP ÷ throughput — to a clean example and to your own data, and name the conditions under which it holds.



# Three clocks on one card

## 1 · One card, three dates, three durations — the convention: a duration = the later date minus the earlier date

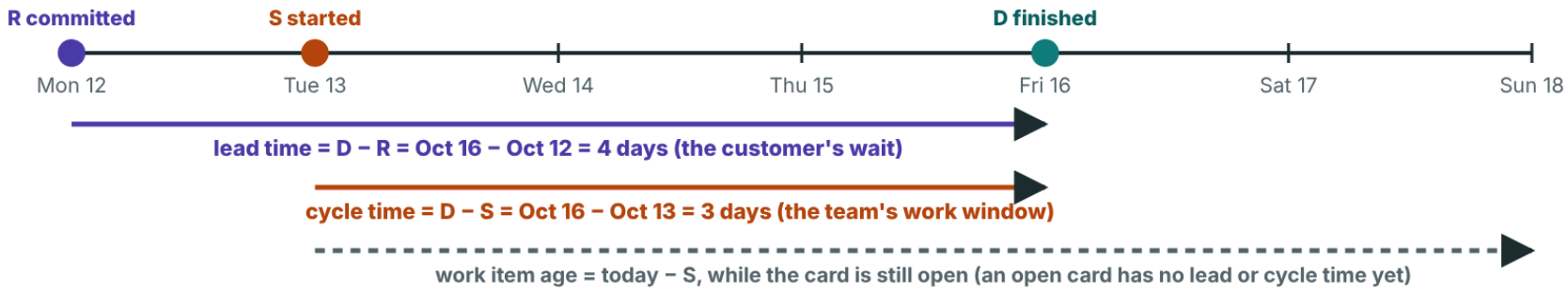
**#2 · 1 pt · sesame shown on every dish**

R committed (entered Ready) Mon Oct 12

S started (entered In progress) Tue Oct 13

D finished (entered Done) Fri Oct 16

**lead 4 d · cycle 3 d · waited 1 d in Ready**




- Lead time: commitment → done — the customer's wait (Kanban University, Atlassian)
- Cycle time: started → finished — the team's work window (Kanban Guide 2025.5)
- Work item age: started → today, for a card still open (Kanban Guide 2025.5) — the measure that acts before it is too late
- Elapsed calendar days, weekends included: the students waited those days too

# The counting convention · always ask 'from when?'

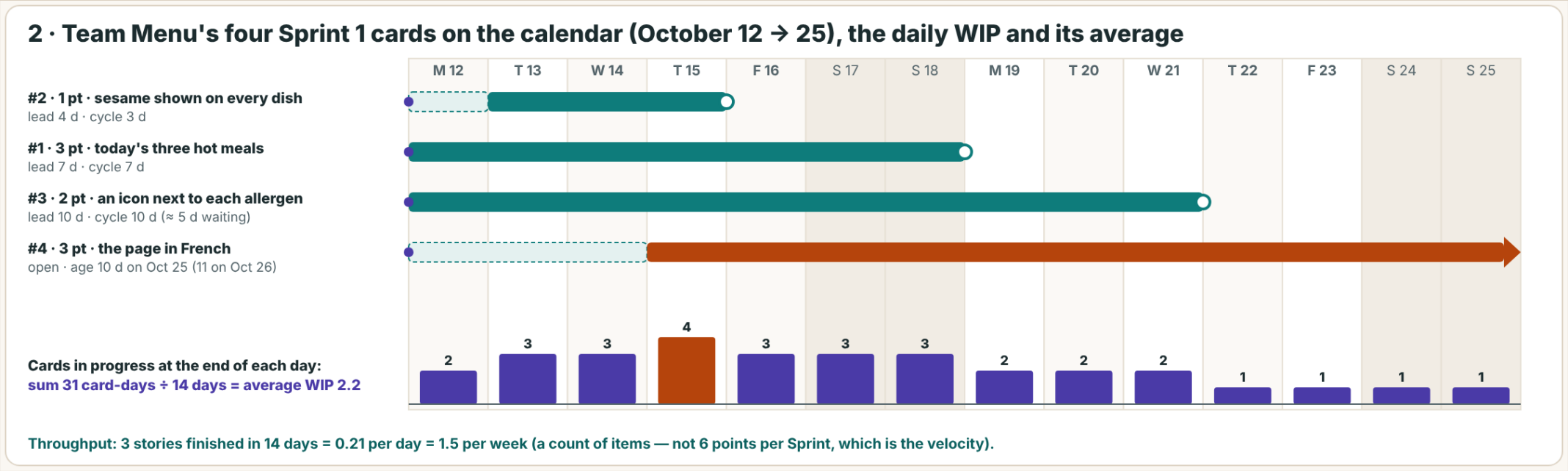
- Course convention: a duration = the later date minus the earlier date (Monday → Friday = 4 days)
- Inclusive counting (Monday → Friday = 5) is acceptable if said aloud; mixing the two in one table is the error
- Story 2: committed Oct 12, started Oct 13, finished Oct 16 → lead 4 d, cycle 3 d, waited 1 d in Ready
- Story 4: started Oct 15, not finished → no lead or cycle time yet; age 10 d on Oct 25, 11 d on Oct 26
- Write the convention at the top of every table you hand in (Lab 04, Mid-Term Part C)

# Team Menu's flow table · Sprint 1 (Oct 12 → 25)

| Story | Pts | Committed | Started | Finished | Lead | Cycle | Age (Oct 25) |
|---|---|---|---|---|---|---|---|
| **2 · sesame on every dish** | 1 | Mon Oct 12 | Tue Oct 13 | Fri Oct 16 | 4 d | 3 d | — |
| **1 · today's three hot meals** | 3 | Mon Oct 12 | Mon Oct 12 | Mon Oct 19 | 7 d | 7 d | — |
| **3 · an icon next to each allergen** | 2 | Mon Oct 12 | Mon Oct 12 | Thu Oct 22 | 10 d | 10 d | — |
| **4 · the page in French** | 3 | Mon Oct 12 | Thu Oct 15 | not finished | — | — | 10 d |

- Story 3's ten days contain about five days of waiting (parked from the 17th) — counted on purpose

# Daily WIP, average WIP, throughput




- Cards in progress at the end of each day: 2 · 3 · 3 · 4 · 3 · 3 · 3 · 2 · 2 · 2 · 1 · 1 · 1 · 1 — 31 card-days ÷ 14 = average WIP 2.2
- Throughput = 3 stories finished ÷ 14 days = 0.21 per day = 1.5 per week — a count of items
- Not 6 points per Sprint: that is the velocity, and it answers a different question (how much, not how many)

# Little's Law (1/2) · the relation among the three measures

## 3 · Little's Law (John D. C. Little, 1961) — the relation among the three measures

**$L = \lambda W$**

L: average number of items in the system (average WIP) · λ: average arrival rate = average departure rate (throughput) · W: average time an item spends in the system.

**Flow form used in this course:**

**average cycle time = average WIP ÷ throughput**

**Units rule: items ÷ (items per day) = days.**

**Class example: 8 items ÷ 2 items/day = 4 days.**

**Team Menu's check (Sprint 1 window)**

**predicted: 2.2 ÷ 0.21 ≈ 10.3 days**

**observed: (3 + 7 + 10) ÷ 3 ≈ 6.7 days**

The gap is not an arithmetic error: the window opens with 0 cards in progress and closes with 1 (story 4 entered and did not leave), the sample is four cards, and the age of the open work grew. Recompute over 28 days on November 8.

**When the law holds (Vacanti, 2015) — five conditions**

1 average arrival rate = average departure rate

2 every item that enters eventually leaves

3 WIP is about the same at the start and the end of the window

4 the average age of the open items neither grows nor shrinks

5 one consistent unit for all three measures

**Part C of the Mid-Term (Nov 16) asks for this table, this line, and a WIP-limit recommendation.**

- John D. C. Little, 1961 (Operations Research 9(3)): L = λ W — items in the system = arrival rate × time in the system
- Flow form: average cycle time = average WIP ÷ throughput · units: items ÷ (items per day) = days
- Class example: 8 items in progress on average, 2 finished per day → average cycle time 4 days

# Little's Law (2/2) · Team Menu's check, and the five conditions

- Predicted: 2.2 ÷ 0.21 ≈ 10.3 days · Observed average cycle time of the finished stories: (3 + 7 + 10) ÷ 3 ≈ 6.7 days
- Not an arithmetic error — the conditions are not met over this window (Vacanti, 2015):
- 1 average arrival rate = average departure rate · 2 every item that enters eventually leaves · 3 WIP about the same at the start and the end
- 4 the average age of the open items neither grows nor shrinks · 5 one consistent unit
- Team Menu: the window opens with 0 cards in progress and closes with 1 (story 4 never left); four cards; the open work aged — recompute over 28 days on November 8
- The Lab 04 line: both numbers and the reason, in your own words

# Activity 3 · fun-quiz ×3, computational (5 minutes)

ACTIVITY · EVERYONE TAKES PART

- One minute each, on your phone or with fingers; write the formula before the number
- 1  A board holds 12 items in progress on average; the team finishes 3 items per day. Average cycle time?
- 2  A team finishes 5 items per week with an average cycle time of 2 weeks. Average WIP?
- 3  A ticket is committed on Monday, started on Wednesday, finished on Friday. Lead time and cycle time, in this course's convention?
- Taking part earns the activity points; the score is for you only

# Example 3 · the flow metrics calculator




- Four cards drawn on the October calendar; lead, cycle and age computed with one days() function that holds the convention
- Daily WIP counted and drawn, average WIP, throughput, Little's Law prediction versus the observed average, the three fun-quiz cases
- Give story 4 a finished date and restart: the gap to the prediction shrinks

4

# The cumulative flow diagram, and Scrum vs. Kanban vs. Scrumban

0:47 – 0:57 · then Activity 4 (0:57 – 1:00)

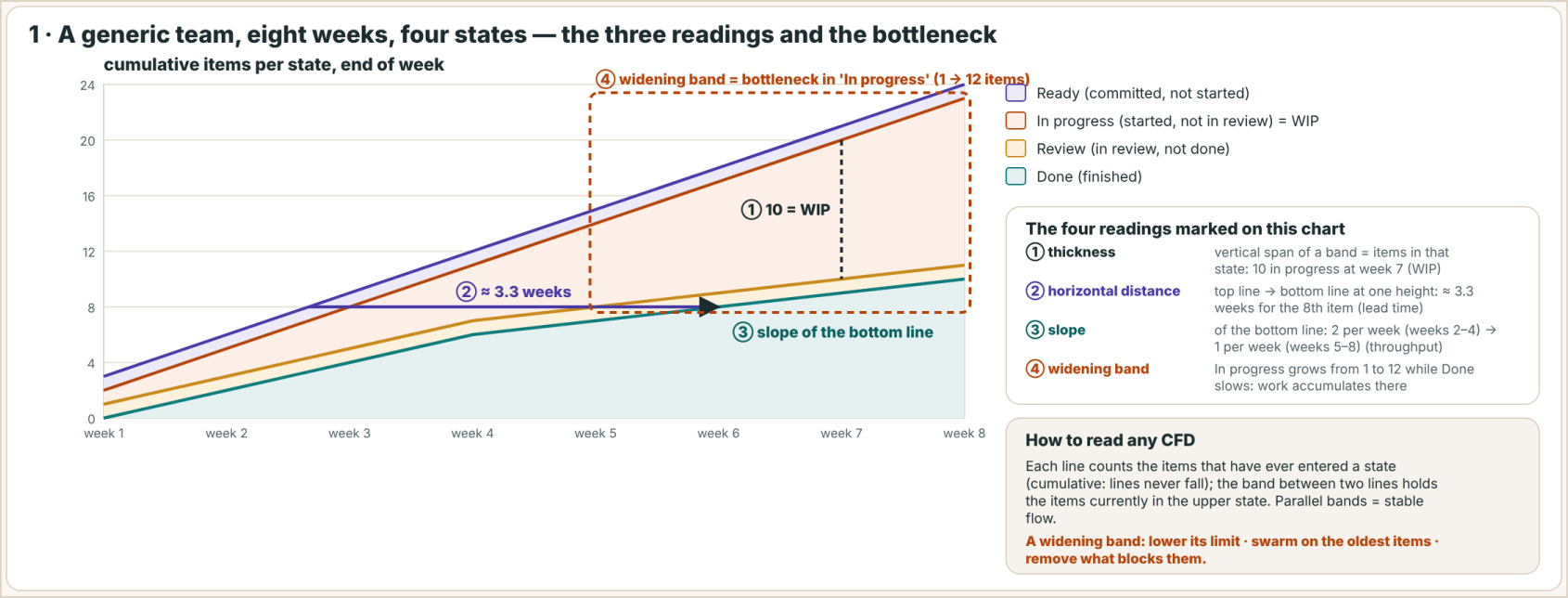
**BY THE END OF THIS LESSON YOU CAN**

Read a cumulative flow diagram: the thickness of a band is the WIP of that state, the horizontal distance between the top and bottom lines is the approximate lead time, the slope of the bottom line is the throughput, and a band that widens is the state where work accumulates.

Write a Service Level Expectation with its two parts — a period and a probability — from historical cycle times, and say why a three-card sample makes it provisional.

Compare Scrum, Kanban and Scrumban on roles, events, artifacts, time-boxes, how work enters, how WIP is controlled and what is measured, and choose one for a given kind of work in five sentences with reasons.

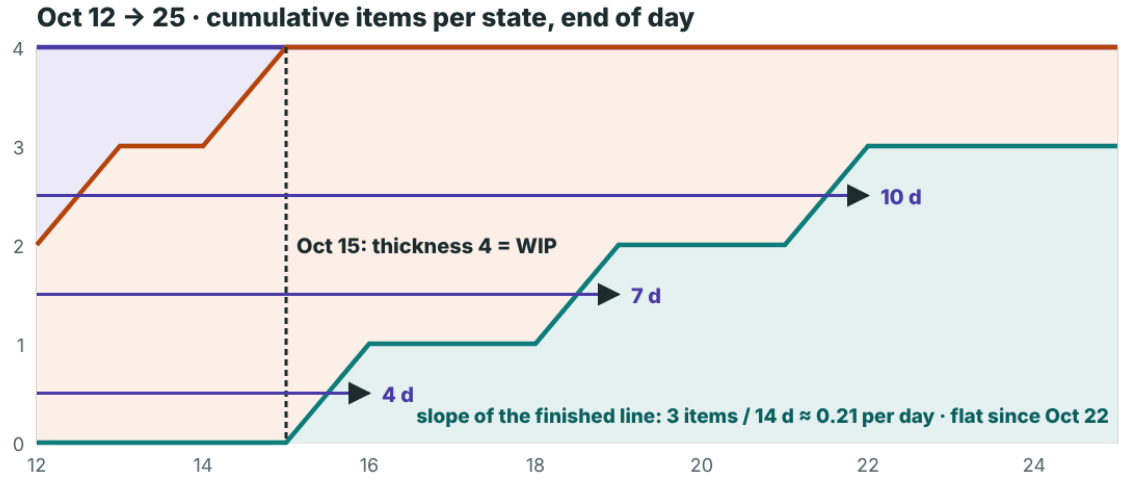
# The cumulative flow diagram (1/2) · how it is built and read




- Each line counts the items that have ever entered a state — cumulative, so lines never fall; the band between two lines = items currently in the upper state
- Thickness = WIP in that state · horizontal distance from top line to bottom line = approximate lead time · slope of the bottom line = throughput
- A band that keeps widening is the state where work accumulates: here In progress grows from 1 to 12 while 'done' slows from 2 to 1 per week

# The cumulative flow diagram (2/2) · Team Menu's Sprint 1

## 2 · Team Menu's Sprint 1 diagram (four cards, fourteen days) and its provisional Service Level Expectation




**SLE (Service Level Expectation) — two parts: a period and a probability**

Built from the team's own historical cycle times, not from a wish.
Team Menu's cycle times so far: 3, 7 and 10 days — three values, so no percentile means much yet.

**"A story should be finished within 10 days of being started, about 85 % of the time." — provisional; recompute on November 8 from the Sprint 1 + Sprint 2 cards.**

Use on October 26: story 4 is eleven days old, above the expectation — that is where the pairing goes first, before any new card is pulled (the Kanban Guide's work item age, the measure that acts before it is too late).

- Oct 15: the In progress band is 4 thick — the WIP peak of Lesson 3
- Gaps from the committed line to the finished line: 4, 7 and 10 days — the lead times, read on the picture
- The finished line rises on Oct 16, 19 and 22 and is flat afterwards: 3 items in 14 days ≈ 0.21 per day

# The Service Level Expectation · a period and a probability

- SLE (Service Level Expectation): 'an item should be finished within N days of being started, P % of the time' (Kanban Guide 2025.5)
- Built from the team's own historical cycle times — a percentile of the data, not a wish or a promise from a plan
- Team Menu's cycle times so far: 3, 7 and 10 days — three values, so the SLE is written in pencil: 'within 10 days of being started, about 85 % of the time — provisional; recompute on November 8'
- Its use today: story 4 is eleven days old, above the expectation — that is where the pairing goes first
- An SLE is checked against work item age every day, not against cycle time once a Sprint

# Scrum, Kanban, Scrumban (1/2) · what each prescribes

| Aspect | Scrum (Scrum Guide 2020) | Kanban (Kanban Guide 2025.5 / Method) | Scrumban (Ladas 2008–09) |
|---|---|---|---|
| **what it is** | a framework | a strategy for optimizing flow / a method | Scrum, plus Kanban's practices |
| **roles · events · artifacts** | 3 · 5 · 3 with commitments | none prescribed; a Definition of Workflow | Scrum's, plus a Definition of Workflow |
| **time-box** | a Sprint ≤ 1 month; the Sprint Goal does not change | none — continuous flow | the Sprint is kept |
| **how work enters** | selected at Sprint Planning | pulled on a clear signal of capacity | selected at Planning; pulled within the Sprint from Ready |
| **WIP control** | implicit (the Sprint Backlog) | explicit WIP limits | explicit WIP limits added |
| **what is measured** | velocity, burndown (common practice) | WIP, throughput, work item age, cycle time; SLE | both sets, side by side |

# Scrum, Kanban, Scrumban (2/2) · which, for which work?

- The question is the kind of work, not the quality of the team
- Scrum fits work that can be planned into a goal for a time-box, with stakeholders who gain from a regular rhythm of review — a product built in increments, like yours
- Kanban fits continuous, unpredictable arrivals of independent items that cannot wait for a planning day — support, operations, maintenance (Anderson's first system)
- Scrumban fits a Scrum team whose Sprints hide queues, that starts too much at once, and that wants cycle times and ages next to its velocity
- Team Menu's decision memo, five sentences: keep Scrum · add WIP limits (In progress ≤ 2, Review ≤ 1) · pull from Ready inside the Sprint · read flow metrics at each Review · therefore Scrumban from October 26, reviewed November 8

# Activity 4 · flash-cards ×6 + zen break (3 minutes)

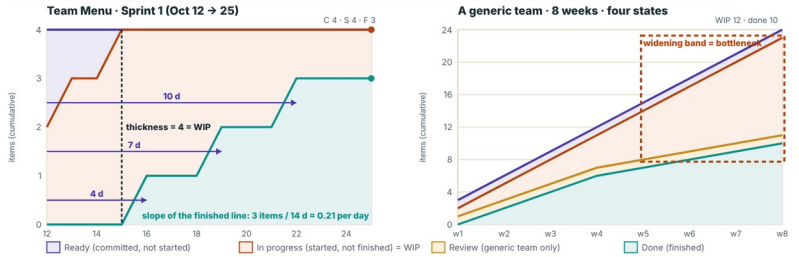
ACTIVITY · EVERYONE TAKES PART

- Six cards on the screen, one at a time: the definition appears first, the term is answered aloud — WIP · lead time · cycle time · throughput · Little's Law · CFD
- Then sixty seconds of silence: eyes on the one card of your own work that is oldest today
- The deck of 24 cards and the formative self-check are in the Study Hub page — not counted toward grades

# Example 4 · CFD builder and method chooser

VISUALIZATION

Cumulative flow diagrams: each line = items that have entered a state · bands = items currently in the state between the lines




**Which method, for which work? · three questions · Team Menu's answers**

**Q1 Can the work be planned into a goal for a time-box, and do the stakeholders gain from a regular rhythm of review?** → keep Scrum
Yes — the cafeteria manager inspects an increment every two weeks; the stories can be planned into a Sprint Goal.

**Q2 Does the work arrive continuously and unpredictably, in independent items that cannot wait for a planning day?** → not a Kanban-only case
No — the backlog is stable between Sprint Plannings; no support stream arrives in between.

**Q3 Do the Sprints hide queues — too much started at once — and does the team want cycle times and ages next to its velocity?** → add Kanban practices
Yes — Sprint 1 ended with four stories started and three finished; story 3 waited five days; story 4 is eleven days old.

**Result: Scrumban — Scrum kept, pull from Ready, WIP limits, flow metrics beside the velocity · from Oct 26, reviewed Nov 8**

(1) Keep Scrum — the work is plannable into a Sprint Goal and the manager inspects an increment every two weeks.
(2) Add WIP limits (In progress ≤ 2, Review ≤ 1) — Sprint 1 ended with four stories started and three finished.
(3) Pull from a Ready column refilled by the Product Owner at the Daily Scrum — nobody starts everything on day one.
(4) Read throughput, cycle time and work item age next to the velocity at each Sprint Review.
(5) Therefore Scrumban from Oct 26, reviewed at the Retrospective of Nov 8.

EXECUTION WINDOW · EVERY STEP IS WRITTEN HERE

```
ages next to its velocity?
    Team Menu: Yes — Sprint 1 ended with four stories
started and three finished; story 3 waited five days; story
4 is eleven days old.  → add Kanban practices
32 | Decision · Q1 yes (Scrum fits the kind of work), Q2 no
(not a continuous stream), Q3 yes (queues and too much
started at once) → Scrumban. Memo:
    (1) We keep Scrum: our work can be planned into a
Sprint Goal and the cafeteria manager wants to inspect an
increment every two weeks.
    (2) We add WIP limits — In progress ≤ 2, Review ≤ 1 —
because Sprint 1 ended with four stories started and three
finished.
    (3) Inside the Sprint we pull from a Ready column that
the Product Owner refills at the Daily Scrum, instead of
everyone starting a story on day one.
    (4) At each Sprint Review we read throughput, cycle
time and the age of the open cards next to the velocity.
    (5) Therefore Scrumban from October 26, to be reviewed
at the Retrospective of November 8.
33 | Done · the left diagram is Lesson 3's table drawn; the
right one is the Companion's drill; the memo is the five
sentences each team writes in the studio. Try: replace CARDS
by your own Sprint 1 cards, or change WEEKS so that the
Review band is the one that widens, and read where the
bottleneck moved.
```

- Team Menu's diagram built one day at a time from three dates per card; the three readings drawn on it
- The eight-week generic diagram with the widening In progress band; the bottleneck named with the three actions
- Three questions about the kind of work → Scrumban, with the five-sentence memo

# The hand-out · a blank flow table for the studio and Lab 04

**Flow table · hand-out (Class 6 studio · Lab 04 part 1 · Mid-Term Part C rehearsal)**

Team: _____ Sprint / window: from _____ to _____ (___ days) Date: _____ Convention: a duration = later date − earlier date (elapsed calendar days)

| Story / ticket | Points | Committed (entered Ready) | Started (entered In progress) | Finished (entered Done) | Lead time (days) | Cycle time (days) | Age today (days) |
|---|---|---|---|---|---|---|---|
| 1. | | | | | | | |
| 2. | | | | | | | |
| 3. | | | | | | | |
| 4. | | | | | | | |
| 5. | | | | | | | |
| 6. | | | | | | | |
| 7. | | | | | | | |
| 8. | | | | | | | |

Lead time = finished − committed · cycle time = finished − started · age = today − started (only for an open card; an open card has no lead or cycle time yet).

**Cards in progress at the end of each day (count the cards started on or before the day and not yet finished)**

| day | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| WIP | | | | | | | | | | | | | | |

**Sum of the daily counts (card-days): _____ ÷ number of days _____ = average WIP _____ items**

Throughput = _____ items finished ÷ _____ days = _____ items per day = _____ items per week

Little's Law, predicted average cycle time = average WIP _____ ÷ throughput _____ = _____ days (units: items ÷ items per day = days)

Observed average cycle time of the finished items = ( _____ + _____ + _____ + … ) ÷ _____ = _____ days

Gap between predicted and observed, and the reason (window, items that entered and did not leave, sample size, growing age): _____

Oldest open card today: #_____, age _____ days · Provisional SLE: "within _____ days of being started, about _____ % of the time"

Recommended WIP limit for In progress: _____ — because _____

**Decision memo (five sentences, each with a reason): Scrum / Kanban / Scrumban — from _____, reviewed on _____**

(1) _____

(2) _____

(3) _____

(4) _____

(5) _____

DIG 123 · Agile Methodologies for Innovation · Class 6 (October 26, 2026) · Diagram 4b · Cyber.SoHo Educational Hub © 2026 — Team Menu is a fictional team (Legal Notice §20)

- Eight rows of tickets, three dates each; lead, cycle and age columns; fourteen daily WIP cells
- Throughput, average WIP, the Little's Law line with its reason, the oldest open card, a provisional SLE, a recommended limit
- The decision memo: five sentences, five reasons, one date

# The four lessons in four sentences

- 1  A kanban is a card that orders work when capacity is free; the board makes the work visible and the pull rule lets the column, not the person, answer 'may I start?'
- 2  WIP is what is started and not finished; the same work finishes with less waiting when less is open at once — so the limit is written on the column as a policy
- 3  Every card carries three clocks — lead time, cycle time, age — throughput counts what comes out, and Little's Law ties them: average cycle time = average WIP ÷ throughput
- 4  The cumulative flow diagram shows all of it at once; an SLE turns history into an expectation; Scrum, Kanban or Scrumban is decided by the kind of work

# Today's studio (after the break)

- 15 min  CQ2 — Checkpoint Quiz 2, individual, closed book, ten items: Scrum events and artifacts, INVEST, story points, one velocity computation, one burndown reading (2 % of the grade)
- 10 min  Sprint 2 Planning — PO: member 3, SM: member 4 of your rotation; Sprint Goal and Sprint Backlog from your Lab 03 stories; a carried-over story keeps its start date
- 25 min  Kanban board set-up — Backlog / Ready / In progress / Review / Done, WIP limits on In progress and Review, a policy per column, commitment and delivery points, 'started' and 'finished' defined once at the top
- 35 min  Flow metrics from Sprint 1 — lead and cycle time of every finished story, age of every open one, throughput per week, average WIP from the daily counts, one Little's Law line (predicted vs observed, one sentence on the gap)
- 20 min  Decision memo — Scrum, Kanban or Scrumban for our product: five sentences, each with a reason (section 4.3)
- 10 min  Sprint Review — one metric and one surprise per team · 5 min  Retrospective — start / stop / continue
- No new lab today; the table and the memo are the first half of Lab 04 (opens November 2)

# Home sprint, and next Monday

- Sprint 2 runs from today to November 8: pull from Ready, respect the limits, write the dates on every card — Sprint 2's dates are your Mid-Term Part C practice data
- Read before Class 7: the Kanban Guide 2025.5 (free, about ten pages — all of it) and Denning, The Age of Agile, Chapter 6 (VitalSource Bookshelf)
- Keep the Daily Scrum lines going (three real lines per week, as in Lab 03); finish the flow table and the Little's Law line started in the studio — they go into Lab 04
- Lab 04 'Flow & Digital Business Brief' opens November 2: metrics table from your Sprint 1 log, one Little's Law line, five blocks of the Business Model Canvas, one MVP hypothesis with its metric; feedback November 8
- Learning Log, five lines · Next Monday, November 2 — Class 7, Agile in the digital business
- Study Hub page: 24 flash-cards and 24 formative questions — not counted toward grades

# Abbreviations used today

| Abbreviation | Meaning | Abbreviation | Meaning |
|---|---|---|---|
| **WIP** | Work In Progress | CFD | Cumulative Flow Diagram |
| **DoW** | Definition of Workflow | SLE | Service Level Expectation |
| **DoD** | Definition of Done | TPS | Toyota Production System |
| **PO / SM** | Product Owner / Scrum Master | KU | Kanban University |
| **CQ2** | Checkpoint Quiz 2 | MVP | Minimum Viable Product |
| **BMC** | Business Model Canvas | APA | American Psychological Association |
| **CC BY-SA** | Creative Commons Attribution-ShareAlike | LMS | Learning Management System (Blackboard) |

# Links and sources (all checked October 23–24, 2026)


- The Kanban Guide 2025.5 — kanbanguides.org/the-kanban-guide/ (CC BY-SA 4.0)
- Kanban University — The Official Guide to the Kanban Method — kanban.university/kanban-guide/
- D. J. Anderson, 'A brief history of Kanban for knowledge work' — djaa.com
- Toyota — the origin of the Toyota Production System — global.toyota
- Little, J. D. C. (1961), 'A Proof for the Queuing Formula: L = λW', Operations Research 9(3) — doi.org/10.1287/opre.9.3.383
- The Scrum Guide 2020 — scrumguides.org · Kanban Guide for Scrum Teams — scrum.org
- Atlassian Agile Coach — WIP limits and Kanban boards — atlassian.com/agile/kanban
- APA — 'Multitasking: switching costs' — apa.org · Modus Cooperandi — Scrumban (Ladas) — moduscooperandi.com
- Full list with one-line explanations: the lecture page, section 'Links'

# Thank you — see you in the studio

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